

LeetCode: 138 Copy List with Random Pointer

By Atul Raut

LeetCode: 138. Copy List with Random Pointer

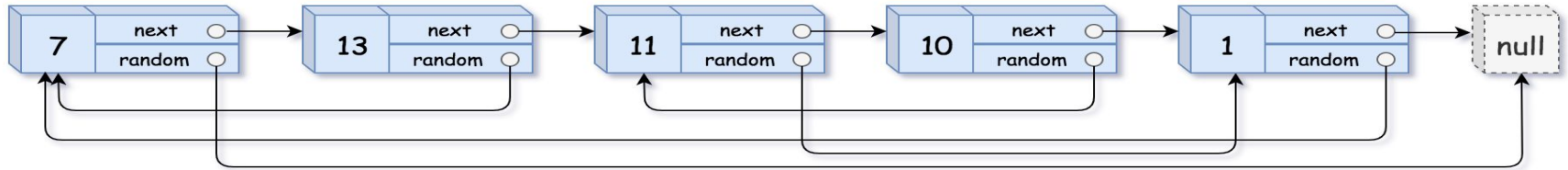
LeetCode Link : [copy-list-with-random-pointer](https://leetcode.com/problems/copy-list-with-random-pointer/)

A linked list is given such that each node contains an additional random pointer which could point to any node in the list or null.

Return a **deep copy** of the list.

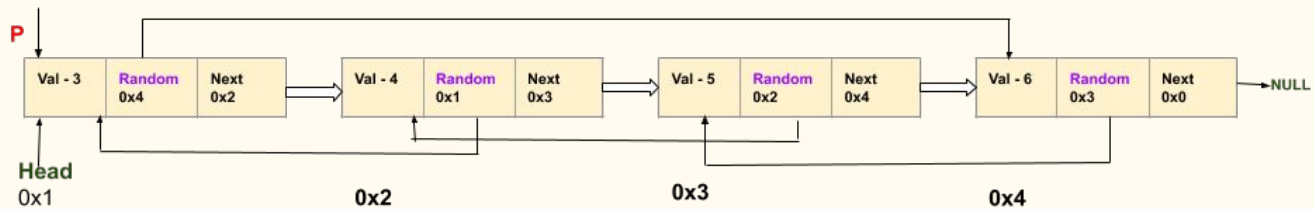
The Linked List is represented in the input/output as a list of n nodes. Each node is represented as a pair of $[\text{val}, \text{random_index}]$ where:

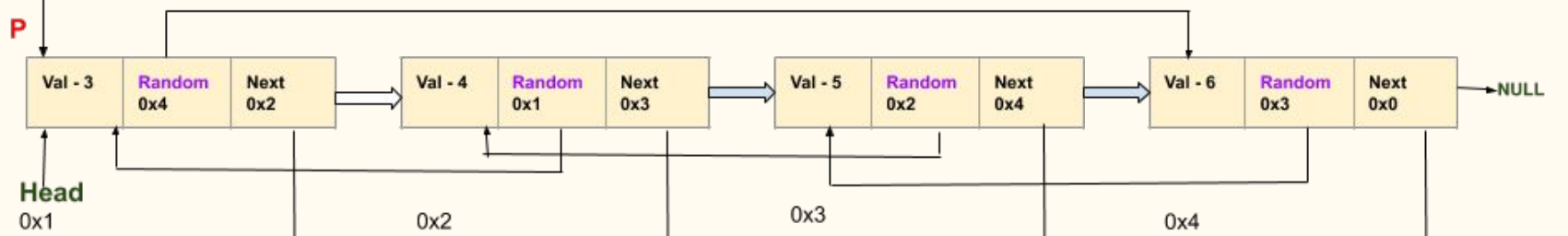
- **val**: an integer representing `Node.val`
- **random_index**: the index of the node (range from 0 to $n-1$) where random pointer points to, or null if it does not point to any node.



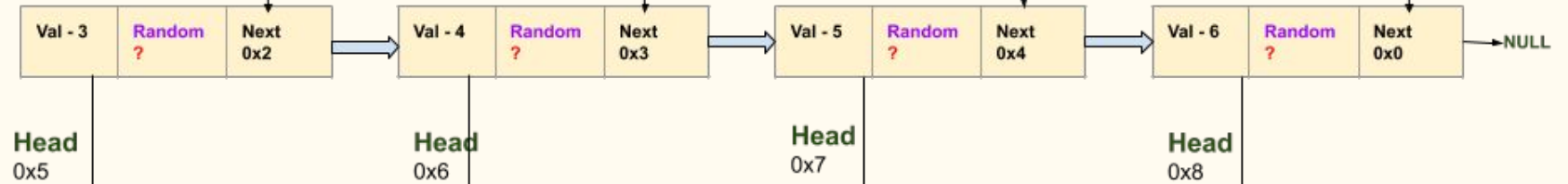
Input: head = `[[7,null],[13,0],[11,4],[10,2],[1,0]]`

Output: `[[7,null],[13,0],[11,4],[10,2],[1,0]]`

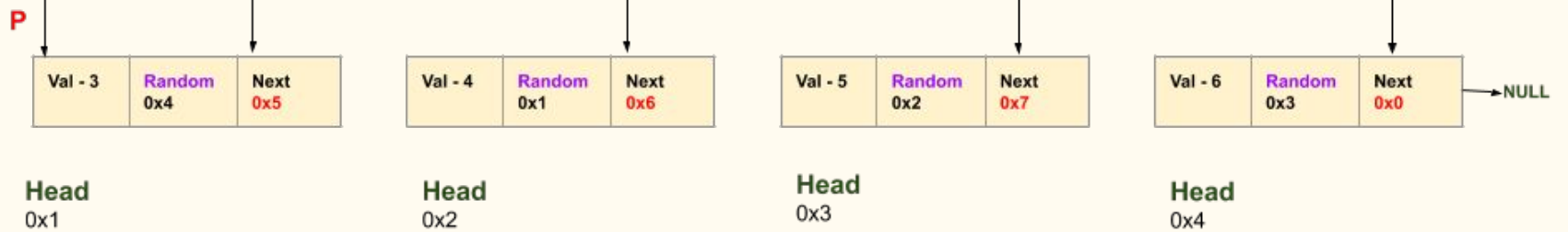




This is Original Linked List



Copied(NN) Linked List



Original Linked List



Head
0x5

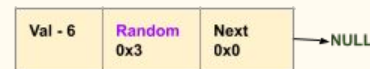
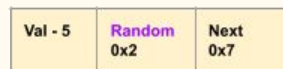
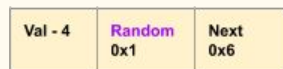
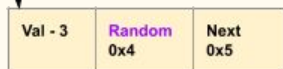
Head
0x6

Head
0x7

Head
0x8

Copied(NN) Linked List

P



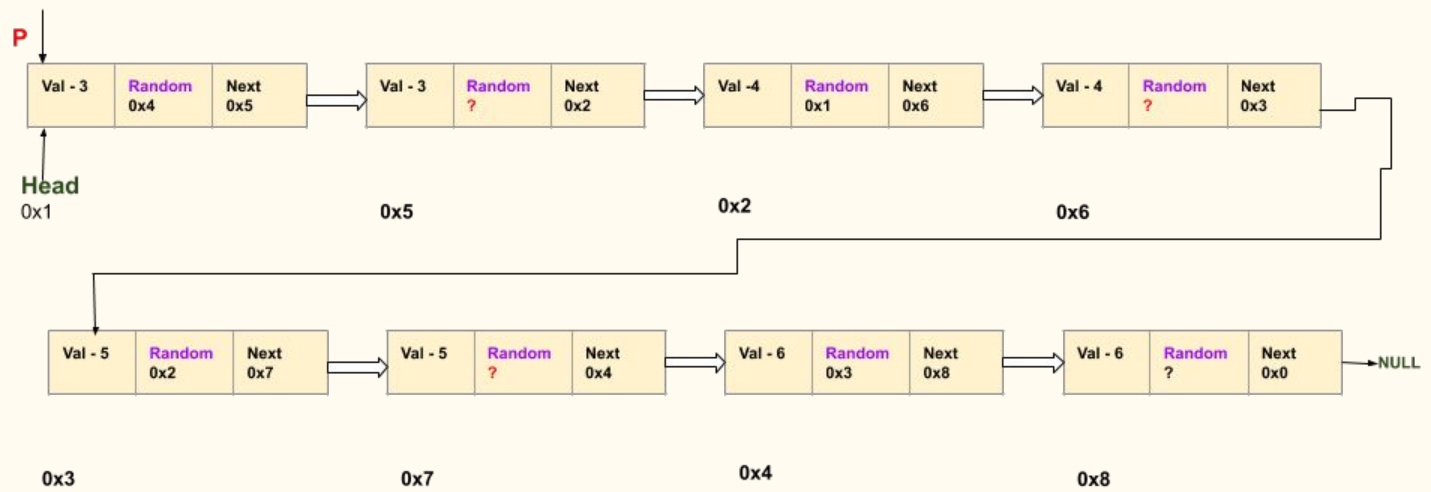
Head
0x1

Head
0x2

Head
0x3

Head
0x4

Original Linked List



Original LinkedList & New Copied Linked List combined Looks like this :



Head
0x5

Head
0x6

Head
0x7

Head
0x8

Copied(NN) Linked List



